RAJALAKSHMI ENGINEERING COLLEGE	**RAJALAKSHMI ENGINEERING COLLEGE** **B.E. COMPUTER SCIENCE AND ENGINEERING** **(CYBER SECURITY)** **CS23533-FOUNDATIONS OF ARTIFICIAL INTELLIGENCE** **III YEAR - V SEM**

TITLE	**"HireAI: A Rule-Based Intelligent Recruitment Expert System with Explainable Decision Support"**
TEAM MEMBERS **REG. NUMBER & NAME**	241901114 & SURVESVARAKUMAR J S 241901067 & NAVANEETHA KRISHNAN E

ABSTRACT:

HireAI: A Rule-Based Intelligent Recruitment Expert System with Explainable Decision Support is a classical Artificial Intelligence application developed to improve the efficiency, transparency, and consistency of the recruitment process. The primary objective of the system is to assist recruiters in screening and evaluating candidates by applying predefined recruitment knowledge rather than subjective human judgment. Candidate profiles are assessed using criteria such as academic performance, technical skills, aptitude, communication ability, and relevant experience to generate intelligent hiring recommendations.

The system employs **IF–THEN production rules** for knowledge representation and utilizes a **forward chaining inference mechanism** to derive conclusions in a logical and explainable manner. Unlike modern recruitment systems that depend on Machine Learning or Deep Learning models, HireAI follows a purely **rule-based expert system** approach, ensuring that every recommendation is accompanied by a clear justification. This makes the system highly transparent, reliable, easy to validate, and simple to customize according to organizational recruitment policies.

The project integrates fundamental concepts from the **Foundations of Artificial Intelligence** syllabus, including **Intelligent Agents, Problem Solving, Knowledge Representation, Rule-Based Expert Systems, Inference Engine, Forward Chaining, and Heuristic Evaluation**. Its key advantages include reduced manual screening effort, consistent candidate evaluation, transparent decision support, low computational complexity, and flexible rule modification. By demonstrating the practical application of classical AI techniques to a real-world recruitment scenario, HireAI highlights the continued relevance and effectiveness of explainable, knowledge-based intelligent systems in decision support.